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1. Which property is *not* required for a distance matrix used in phylogeny?

- A. Symmetry
- B. Non-negativity
- C. Triangle inequality
- D. Positive-definiteness

2. How many edges are in an unrooted binary tree that has 17 leaves?

Answer:

3. Dollo's principle states that:

- A. Evolution is reversible at the molecular level
- B. The same complex organ cannot be regained once lost
- C. All characters evolve at the same rate
- D. The root of a tree must be placed at the midpoint

4. UPGMA always produces the correct tree when the input distance matrix is additive. (True/False)

5. Neighbor-joining guarantees to recover the true tree whenever the distance matrix is additive. (True/-False)

6. For the additive matrix

$$\begin{pmatrix} 0 & 13 & 21 & 22 \\ 13 & 0 & 12 & 13 \\ 21 & 12 & 0 & 13 \\ 22 & 13 & 13 & 0 \end{pmatrix}$$

compute the neighbor-joining matrix value D_{12}^*

Answer:

7. For one character site, the leaves are labeled A,C,C,T. At an internal node v , daughter(v) has minimum scores $s_A = 2$, $s_C = 1$, $s_G = 3$, $s_T = 2$ and son(v) has $s_A = 3$, $s_C = 2$, $s_G = 2$, $s_T = 1$. Compute the minimum possible score $s_k(v)$ when v is labeled C.

Answer:

8. The ultrametric assumption implies that:

- A. All leaves are equidistant from the root
- B. All internal nodes have degree 3
- C. The tree is unrooted
- D. Evolutionary rates vary among lineages

9. LimbLength(i) computed by neighbor-joining can be negative for some leaves. (True/False)

10. For the quartet V1,V2,V3,V4 with $D_{23} = 12$, $D_{12} = 13$, $D_{13} = 21$, $D_{24} = 13$, $D_{34} = 13$, compute the estimated limb length of V2 when V2 and V3 are joined as neighbors.

Answer: